Long Memory & Spectral Properties of Gold Returns

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1 Executive Summary

Gold is often called a "safe haven": an asset that holds its value when other markets are in turmoil. But does that make gold prices predictable? Can past price movements tell us anything useful about where gold is heading next?

This report investigates the statistical structure of gold futures returns using 26 years of daily data (August 2000 to November 2026, 6,570 trading days) and the corresponding monthly returns. The analysis proceeds in three stages: a descriptive look at the return distribution, a nonparametric spectral analysis to detect hidden patterns across frequencies, and a parametric long-memory model to test formally for persistence.

Key Findings at a Glance:

- **Gold returns are essentially unpredictable from past prices alone.** The spectral white noise test does not reject at any of the tested frequency thresholds, and the estimated memory parameter is not significantly different from zero.
- **Gold prices show a strong long-run upward trend**, rising from about 278 to about 4,073 USD per ounce over the sample, with a notable correction in 2011 to 2013.
- **Volatility is not random.** Calm and turbulent periods cluster together. The most volatile years in the sample were 2008, 2026, 2006.
- **No seasonal patterns were found.** Gold does not systematically perform better or worse at any calendar frequency.
- **This is consistent with an efficient market.** In a deep, globally traded market like gold, publicly available information, including past prices, is rapidly priced in and leaves no exploitable edge.

About this Report

Data source	Yahoo Finance, GC=F (gold futures, continuous front-month)
Period	August 2000 to November 2026 (daily & monthly)
Methods	Descriptive analysis, spectral analysis, white noise tests, ARFIMA
Tools	R (`quantmod`, `fracdiff`, `kableExtra`)
Author	Filip Popovic

2 Data

2.1 Source & Preparation

Gold futures prices (continuous front-month contract, ticker GC=F) are sourced from Yahoo Finance via the `quantmod` package. The series is available from August 2000 onwards. Log returns are computed as the first difference of log closing prices: $r_t = \ln(P_t) - \ln(P_{t-1})$. This makes the series scale-invariant and approximately symmetric around zero.

Daily returns are used for the main analysis. To check for seasonal patterns at a lower frequency, a monthly series is built from the last closing price of each month (315 monthly returns). Using month-end prices rather than monthly averages is important: averaging prices within a month would introduce artificial autocorrelation into the monthly returns.

One limitation should be kept in mind: a continuous front-month series is stitched together from consecutive futures contracts, so small artificial jumps can occur on roll dates.

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3 Descriptive Analysis

3.1 Price Level and Daily Returns

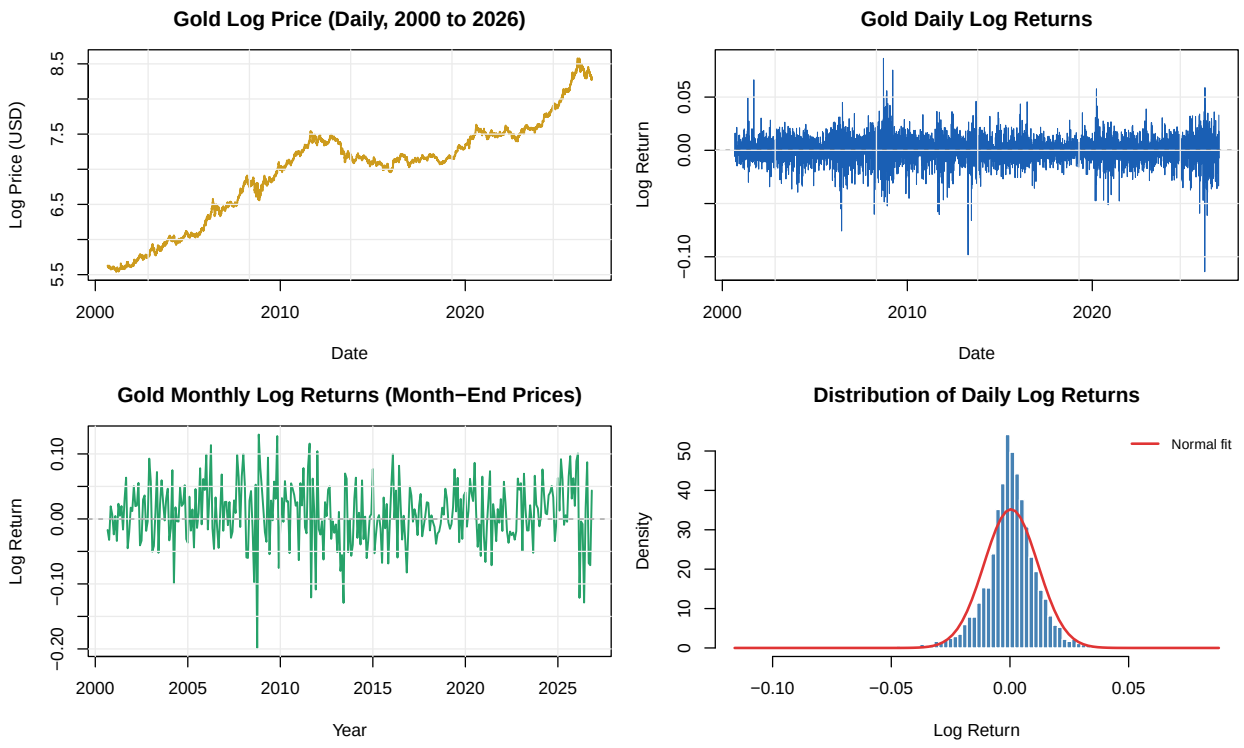


Figure 1: Top row: gold log price (left) and daily log returns (right). Bottom row: monthly log returns (left) and the distribution of daily returns with a fitted normal curve (right). The heavy tails of the distribution show that extreme price moves occur more often than a normal distribution would predict.

The log price series (top left) shows a persistent upward trend, with a strong rise until 2011, a correction between 2011 and 2013, and renewed strength from 2019 onwards that accelerated sharply in the most recent years. This non-stationarity in the price level is the reason why the rest of the analysis works with log returns.

The daily log returns (top right) fluctuate around zero without a visible trend, but they show pronounced volatility clustering: calm periods alternate with turbulent episodes. The return distribution (bottom right) is roughly bell-shaped, but with much heavier tails than the fitted normal curve (excess kurtosis of 6.2), a well-known feature of financial returns.

3.2 Volatility Structure

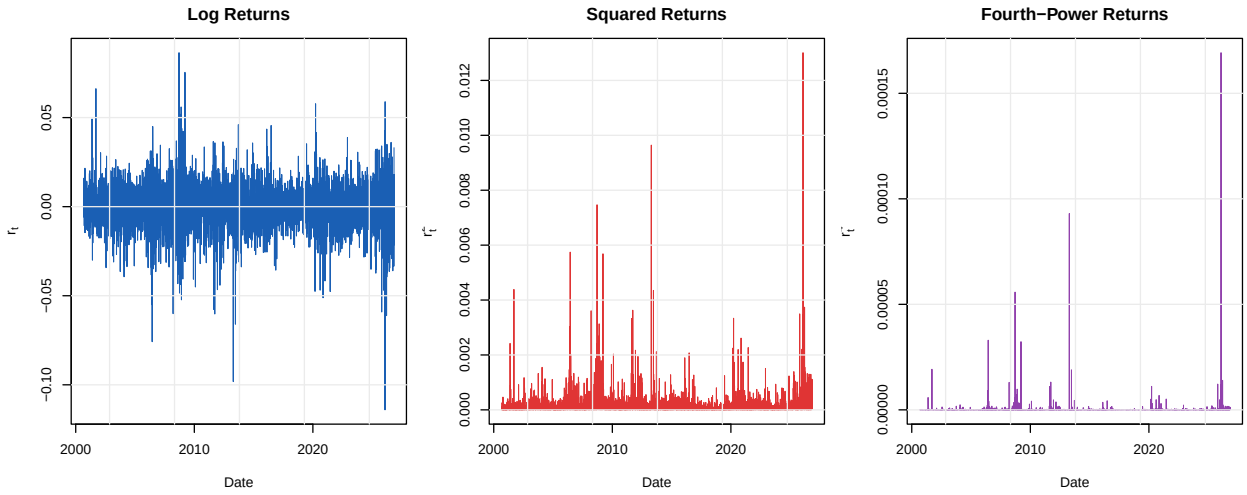


Figure 2: Daily log returns (left), squared returns (centre) and fourth-power returns (right). The clustering in the squared and fourth-power series is the statistical fingerprint of time-varying volatility: calm periods tend to be followed by calm periods, and turbulent periods by turbulent ones.

Plotting squared and fourth-power returns is a standard diagnostic for time-varying volatility. For a series with constant variance, all three panels would look like structureless noise. Instead, the squared returns (centre) show large values in groups, which means that periods of high volatility persist over time. This is the typical pattern behind ARCH/GARCH models. The fourth-power series (right) singles out the most extreme days. The three largest daily moves in the sample were -11.4% on 30 January 2026, -9.8% on 15 April 2013, 8.6% on 17 September 2008. So while the **level** of returns may be close to white noise, the **volatility** clearly is not.

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4 Spectral Analysis

Spectral analysis asks: *at which frequencies does most of the variance in gold returns occur?* A white noise process spreads its variance evenly across all frequencies and produces a flat periodogram. A systematic pattern, such as a seasonal cycle or a persistent trend, would show up as a clear peak. Following the lecture, frequencies are measured in radians, $0 < \lambda \leq \pi$, and the periodogram is $I(\lambda) = \text{spec}/(2\pi)$.

4.1 Periodogram of Daily Gold Returns

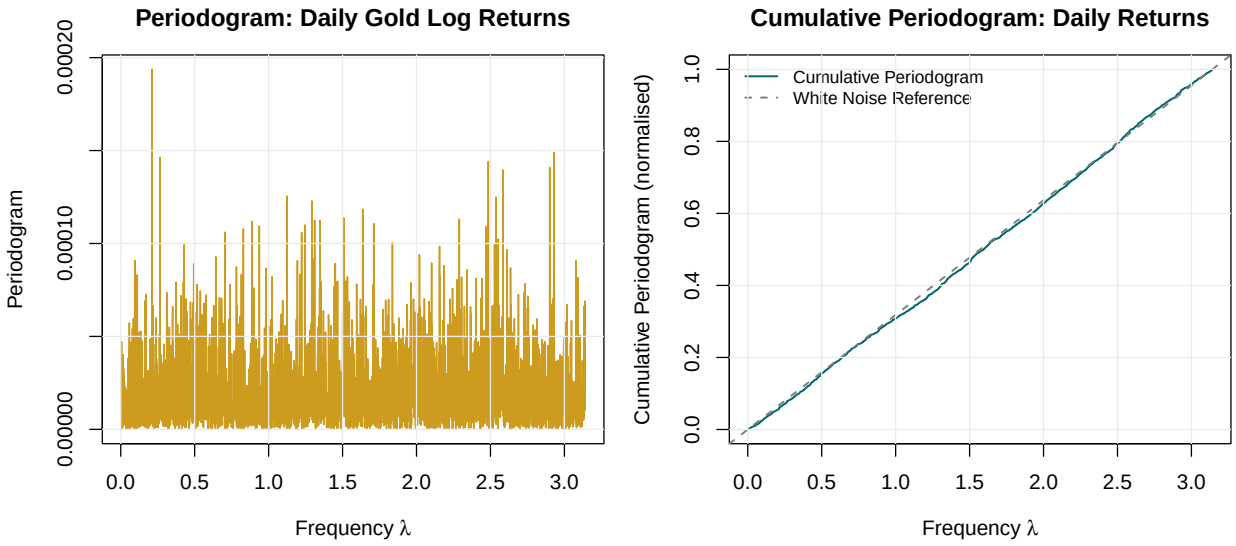


Figure 3: Periodogram (left) and cumulative periodogram (right) of daily gold log returns. A flat periodogram and a cumulative periodogram that follows the diagonal white noise line are the signature of a serially uncorrelated process.

The periodogram of daily gold returns shows no dominant peak and no low-frequency bulge. The cumulative periodogram follows the diagonal reference line closely, with only small random deviations. Already visually, there is little sign of linear structure in daily gold returns.

4.2 White Noise Test: Daily Returns

Table 1: White noise test for daily gold log returns: sum of the periodogram at the j lowest frequencies times $4\pi/s^2$, compared with the 95% quantile of the χ^2_{2j} distribution.

j	Frequency (j)	Test Statistic	Chi-sq Quantile (95%)	p-value
1000	0.9563	1951.79	2105.15	0.7757
2000	1.9127	3922.59	4148.25	0.8061
3000	2.8690	6034.91	6181.31	0.3729

The test checks whether too much of the variance is concentrated at low frequencies, which would indicate persistence. Under the null hypothesis of white noise, the statistic follows approximately a χ^2 distribution with $2j$ degrees of freedom. Across all 3 thresholds the p-values range from 0.373 to 0.806. **The white noise hypothesis is not rejected at any tested threshold.**

As a robustness check, a Ljung-Box test with 20 lags gives a p-value of 0.028. This small rejection should be read with care: the Ljung-Box test is known to reject too often when volatility clusters, which is clearly the case here. Even if a tiny amount of short-term autocorrelation exists, it is far too small to be exploited after trading costs.

4.3 Seasonal Patterns in Monthly Returns

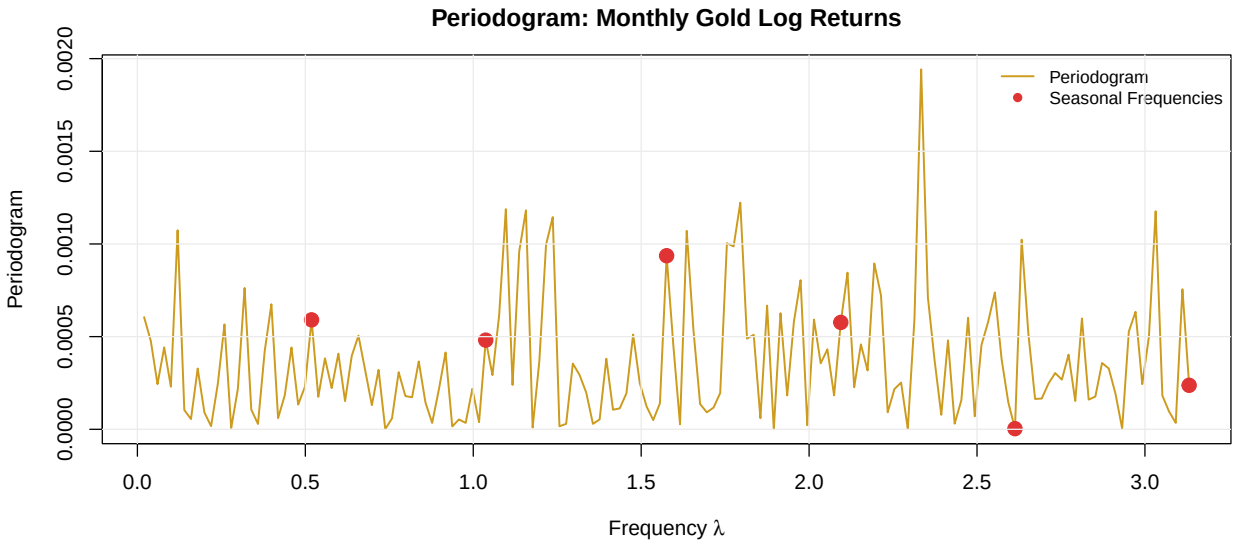


Figure 4: Periodogram of monthly gold log returns. Red points mark the seasonal frequencies $\lambda_k = 2\pi k/12$, $k = 1, \ldots, 6$ (cycles of 12, 6, 4, 3, 2.4 and 2 months).

The red points mark the six seasonal frequencies, where a calendar pattern (annual, semi-annual, quarterly and so on) would produce a peak. None of them stands out from the rest of the periodogram.

4.4 Seasonal Pattern Test

Table 2: Seasonal pattern test for monthly gold log returns: sum of the periodogram at the first j seasonal frequencies times $4\pi/s^2$.

Seasonal Frequencies (j)	Test Statistic	Degrees of Freedom	p-value
1	3.219	2	0.2000
3	10.946	6	0.0900
6	15.403	12	0.2201

The seasonal test focuses on the calendar frequencies only, which makes it more sensitive to periodic patterns than a full-spectrum test. The p-values lie between 0.090 and 0.220. **Gold returns show no statistically significant seasonal pattern**, which is consistent with a deep, globally liquid market that is not driven by calendar effects.

5 ARFIMA Long-Memory Model

5.1 Model Selection

Table 3: ARFIMA model comparison for daily gold log returns. The bold row marks the specification preferred by AIC.

Model	d	Log-Likelihood	Parameters	AIC
ARFIMA(0,d,0)	**4.58e-05**	**20116.21**	**2**	**-40228.42**
ARFIMA(1,d,0)	4.58e-05	20116.34	3	-40226.68
ARFIMA(0,d,1)	4.58e-05	20116.35	3	-40226.70
ARFIMA(1,d,1)	7.26e-05	20116.36	4	-40224.72

The ARFIMA model measures long-run memory through the parameter d. A value of $d = 0$ means no memory, positive values indicate persistence and negative values indicate anti-persistence. Four specifications are estimated by maximum likelihood with `fracdiff` and compared by AIC (lower is better). The preferred model is **ARFIMA(0,d,0)**. The simplest specification wins, so the data give no reason to add AR or MA terms.

All four estimates of d lie at essentially zero. This needs one caveat: fracdiff only searches for d in the range from 0 to 0.5, so an estimate at the lower bound cannot distinguish between no memory and anti-persistence. To test $d = 0$ properly, a semiparametric estimator (the smoothed periodogram estimator `fdSperio`), which allows negative values and provides a standard error, is used as a check. It gives $\hat{d} = 0.019$ with a standard error of 0.027 (p-value 0.468). **The memory parameter is not significantly different from zero: daily gold returns show no long memory.**

5.2 Residual Diagnostics

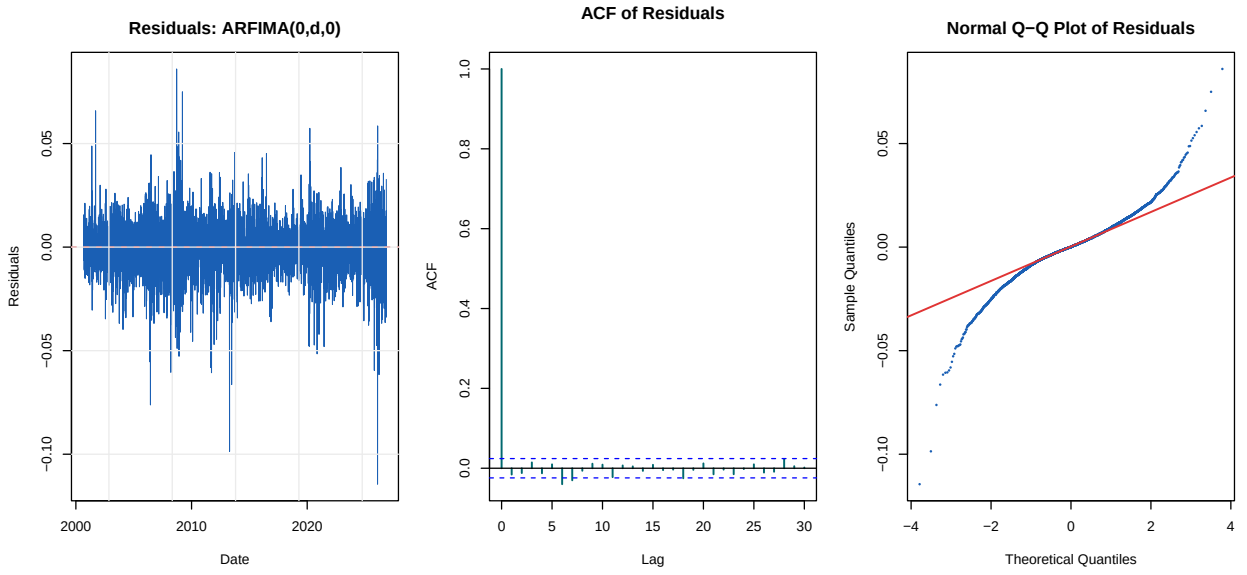


Figure 5: Diagnostic plots for the preferred ARFIMA model: residuals over time (left), autocorrelation function of the residuals with 95% bands (centre) and normal Q-Q plot (right).

Because d is close to zero, the residuals look almost identical to the original returns. In the ACF, 3 of 30 autocorrelations fall outside the 95% bands, while about 1.5 would be expected by chance alone. All of them are very small in absolute size (at most 0.04), which matches the borderline Ljung-Box result: there may

be a trace of short-term dependence, but it is economically negligible. The Q-Q plot shows clear deviations in both tails: the residuals inherit the heavy tails of gold returns and are far from normal at the extremes. This does not affect the conclusion about linear dependence, but it is another sign that the volatility of gold returns needs its own model.

5.3 Parametric vs. Nonparametric Spectrum

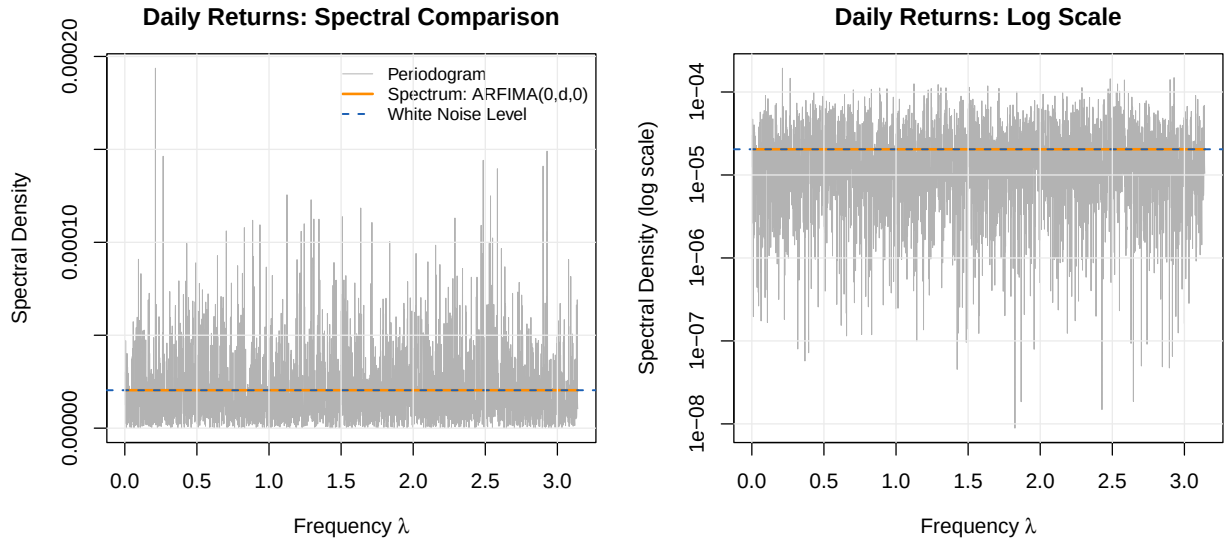


Figure 6: Periodogram of daily gold returns (grey) compared with the spectral density implied by the preferred ARFIMA model (orange) and the white noise level (blue, dashed), on a linear scale (left) and a log scale (right).

The spectral density implied by the ARFIMA model (orange) is flat and lies on top of the white noise level (blue). A long-memory process would instead show a spectrum that rises steeply towards frequency zero. The periodogram scatters randomly around this flat line on both scales, so the parametric model has no structure to explain beyond white noise.

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6 Conclusion

The spectral and long-memory analysis of daily gold returns delivers a consistent message.

The spectral white noise test does not reject at any tested threshold, the monthly returns show no seasonal pattern, and the memory parameter d is not significantly different from zero. In plain terms: **past gold prices contain essentially no usable linear information about future gold prices.**

This finding is in line with the weak form of the Efficient Market Hypothesis. Gold is one of the most liquid and globally traded markets in the world. Any predictable pattern based on past prices would be quickly identified and traded away. The result is therefore not a weakness of the analysis but the expected outcome for a well-functioning market.

One caveat deserves emphasis: white noise in returns does not mean white noise in risk. The clustering in the squared returns shows that **the size of price swings is predictable even when their direction is not.** This distinction is central to financial risk management. Models for time-varying volatility, such as GARCH, would be the natural next step.

This analysis was created by Filip Popovic as part of a data analyst portfolio. For questions or inquiries, ***contact me at fpopovic01@gmail.com****.*